The BFKL terms of the projectile:
three-ρ cancellation, two-ρ terms, colour-singlet form

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1 Result

Your file. `bfkl.tex` is, term by term, the list of log $\frac{\vee}{k^+}$ terms I sent (`Finite_terms_split.pdf`, Secs. 2–4). The only differences are in writing: Group I Row II part 1 has $\frac{g^4 N_c}{16\pi^4}$ in place of $2 \times \frac{g^4 N_c}{32\pi^4}$, and in Group I Row III d^3N_1 the sign is in front. Every row equals the coefficient of log $\frac{\vee}{k^+}$ of the same row of `Finite_terms_1.tex`, pointwise in the transverse coordinates (Table 1).

As it stands, the list does not give BFKL. Two contributions are missing. They are missing in the rows of `Finite_terms_1.tex` as well, so this list (and my earlier notes built from it, Sec. 8) inherited them:

(A) **Group I Row III.** The term with $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$ (first term of d^3N_{3a}) is what is left of the edge kernel of part 2. It is not integrable at $z \to w$. It must be replaced by the $\mathcal{D}$ term (fix F3). Your `Evolution_terms.tex` already has $\mathcal{D}$ in Group I Row III, Sec. 3.1.

(B) **Group III Rows III (both parts), V, VI.** Their p^+ integrals carry the phase $e^{-ik\cdot(y'-z)p^+/k^+}$. After the y' integral, the region $y' \to z$ gives log $\frac{\vee}{k^+}$ terms in which the measured gluon of the conjugate amplitude is emitted by the soft gluon. This is the same mechanism that gave H_4 in Group II Row IV, Sec. 3.2.

Both are added in red/blue in `bfkl_corrected.pdf` and in Sec. 3.

The test. All terms are transcribed into the reordering engine used for `Evolution_terms.tex`, with the ρ's in your order and every c.c.:

	as in `bfkl.tex`	with (A)	with (A) and (B)
symmetrized three-ρ terms, $\lvert\text{sum}\rvert/\sum\lvert\text{terms}\rvert$	7.9×10^{-4}–4.5×10^{-2}	$\leq 4.8 \times 10^{-17}$	$\leq 3.7 \times 10^{-17}$
two-ρ kernel structures that differ from $\mathcal{B} - \mathcal{J}$	33 of 75	9 of 67	0 of 67

With (A) and (B):

- **The three-ρ terms cancel**, analytically (Sec. 4, every row checked separately) and numerically to 10^{-17}.
- **The two-ρ terms give BFKL⊗LO − JIMWLK⊗LO** in every one of the 67 kernel structures (Sec. 5):

$$\boxed{\sum_{\text{all rows}} \frac{d^3N}{d^3k} = \mathcal{N}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\mathcal{B}-\mathcal{J}\Big], \qquad \mathcal{N}\equiv\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}.} \tag{1.1}$$

  Here $\mathcal{B}$ is the bracket of your Sec. 4 (BFKL contribution, the same as App. E, Eq. (LOBFKL)) and $\mathcal{J}$ the bracket of your Sec. 3 (JIMWLK contribution) of `Evolution_terms.tex`.

- **The BFKL term.** With the $\mathcal{N}\log\frac{\vee}{\Lambda}\int\mathcal{J}$ of `Evolution_terms.tex`:

$$\boxed{\frac{d^3N_{\rm LL}}{d^3k} = \mathcal{N}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\left[\log\left[\frac{k^+}{\Lambda}\right]\mathcal{J}+\log\left[\frac{\vee}{k^+}\right]\mathcal{B}\right].} \tag{1.2}$$

  JIMWLK (target) multiplies $\log\frac{k^+}{\Lambda}$ and BFKL (projectile) multiplies $\log\frac{\vee}{k^+}$. $\mathcal{B}$ is written out in Sec. 6, and in colour-singlet form in Sec. 7.

2 Notation and the check of your rows

- $\mathcal{N}=\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$, $\mathcal{K}(X)=\frac{X}{X^2}$, $K(a,b,z)=\mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $K(a,a,z)=\frac{1}{(a-z)^2}$, $U^{bc}=U^{cb}$.
- In Secs. 3–4: $X'=\mathcal{K}(x'-w')$, $X_w=\mathcal{K}(x-w)$, $X_z=\mathcal{K}(x-z)$, $Y=\mathcal{K}(y-z)$, $Y_w=\mathcal{K}(y-w)$, $Z_w=\mathcal{K}(z-w)$, $\mathcal{C}^a_L=\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')$, $p_8=\frac{ig^4}{8\pi^5k^+}$.
- **Transcription.** Every row is transcribed twice:
  - numerically (`../bfkl_check/corrected_lvk.py`), and compared with the exact coefficient of $\log\frac{\vee}{k^+}$ of the same row of `Finite_terms_1.tex`;
  - into the reordering engine (`rows.py`), and compared with the numerical transcription (`validate.py`).

Table 1: Your rows. Column 2: your term vs. the coefficient of $\log\frac{\vee}{k^+}$ of the same row of `Finite_terms_1.tex`, pointwise in all transverse coordinates (for Group III Rows III, V, VI: before the y' integral, i.e. without (B)). Column 3: engine transcription vs. numerical transcription (classical charges, three random configurations).

row	yours vs. exact	engine vs. numerical
Group I Row II 2a	0	4.8×10^{-14}
Group I Row III	1.1×10^{-14}	2.4×10^{-15}
Group I Row IV	8.1×10^{-16}	7.8×10^{-16}
Group I Row V part 1	0	3.2×10^{-16}
Group I Row V 2a	3.1×10^{-16}	1.2×10^{-15}
Group I Row V 2b finite	6.3×10^{-16}	8.9×10^{-16}
Group II Row I part 1	2.0×10^{-15}	4.1×10^{-15}
Group II Row I part 2	2.2×10^{-16}	1.4×10^{-13}
Group III Row I	9.0×10^{-16}	5.3×10^{-15}
Group III Row II	8.8×10^{-16}	8.5×10^{-15}
Group III Row III part 1	3.2×10^{-16}	1.0×10^{-14}
Group III Row III part 2	6.1×10^{-16}	1.9×10^{-15}
Group III Row V	3.5×10^{-16}	1.5×10^{-15}
Group III Row VI	2.3×10^{-16}	4.0×10^{-15}
Group I Row III with (A)	—	1.3×10^{-15}

Group I Row II part 1, Row V 2b UV and Group II Row IV contain $\frac{1}{\epsilon}$ and are compared analytically (Sec. 5.2). In Group I Row III (as written) the engine uses the simplified kernel of Eq. (3.1).

3 The two changes

3.1 (A) Group I Row III: the edge kernel and $\mathcal{D}$

Your three equations in one line. With $O_{\rm III} \equiv f^{abd}\mathcal{C}^a_L\big[U^{de}(x)U^{bc}(z) - U^{de}(x)U^{bc}(y)\big]\rho^e(x)\rho^c(y)$ (charges in your order), the terms of your d^3N_1, d^3N_2, d^3N_{3a} are, exactly (1.1×10^{-15}, `checks.py`):

$$\begin{aligned}
d^3N_1 &= \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\left(-\,p_8\log\left[\frac{\vee}{k^+}\right]\right)(X'\cdot X_w)\,K(x,y,z)\,O_{\rm III},\\
d^3N_2 + d^3N_{3a}\big|_{\text{terms 2, 3}} &= \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\left(+\,p_8\log\left[\frac{\vee}{k^+}\right]\right)(X'\cdot Z_w)\,K(x,y,z)\,O_{\rm III},\\
d^3N_{3a}\big|_{\text{term 1}} &= \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\left(+\,p_8\log\left[\frac{\vee}{k^+}\right]\right)(X'\cdot Y)\,K(w,w,z)\,O_{\rm III}.
\end{aligned}\tag{3.1}$$

The six terms with $\frac{1}{(x-w)^2-(x-z)^2}$ and the two remaining terms of d^3N_{3a} add up to the single kernel $(X'\cdot Z_w)K(x,y,z)$.

The problem. The last line has $K(w,w,z)=\frac{1}{(z-w)^2}$ and $O_{\rm III}$ does not vanish at $z\to w$, so $\int_z$ diverges. It is the $\log\frac{\vee}{k^+}$ part of the edge kernel $\frac{(x-z)^2+(x-w)^2}{[(x-z)^2-(x-w)^2]^2}$ of part 2, through $\log\frac{k^+-\Lambda}{\vee} = -\log\frac{\vee}{k^+}+\dots$. It has no partner in any other row: with it, the three-ρ terms do not cancel (Sec. 4).

The change. Fix F3 replaces the edge kernel by

$$\mathcal{D} = \frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\,p_8\,(X'\cdot Y_w)\,K(x,w,z)\,f^{abd}\mathcal{C}^a_L\big[U^{de}(x)U^{bc}(w) - U^{de}(x)U^{bc}(y)\big]\rho^e(x)\rho^c(y),\tag{3.2}$$

i.e. $O_{\rm III}$ with $U^{bc}(z)\to U^{bc}(w)$: the charge y emits the measured gluon w, and the soft gluon z is exchanged between the measured gluon and the rotated charge $U(x)\rho(x)$.

- Your `Evolution_terms.tex` already has it with $\log\frac{\vee}{\Lambda}$: the $\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$ terms of Group I Row III d^3N_2 are the two-ρ terms of $\mathcal{D}$.
- This region multiplies $\log\frac{k^+}{\Lambda} = \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+}$. So the $\log\frac{\vee}{k^+}$ list gets $-\mathcal{D}$, which is $+p_8(X'\cdot Y_w)K(x,w,z)$ in the convention of Eq. (3.1).

In your format (red: remove, blue: add), d^3N_{3a} becomes

$$\begin{aligned}
\frac{d^3N_{3a\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{
&-\frac{ig^4f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\
&-\frac{ig^4f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2\,(y-w)^2\,(x-z)^2\,(w-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(w)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\
&+\big(\text{your second and third terms, unchanged}\big)\Bigg\}.
\end{aligned}\tag{3.3}$$

Note for `Finite_terms_1.tex`. The other edge-kernel terms of Row III (the $\log(1-\frac{k^+}{\vee})$ term of d^3N_1, the $\log\frac{(x-w)^2k^+-(x-z)^2(k^+-\Lambda)}{(x-w)^2k^+-(x-z)^2\Lambda}$ and $\mathcal{A}$ terms of d^3N_{3a}, d^3N_{3b}) come from the same edge kernel. Their finite parts should follow from your derivation of $\mathcal{D}$; this is open.

3.2 (B) Group III Rows III, V, VI: the region $y'\to z$

The rows. In `Finite_terms_1.tex`, Group III Rows III, V, VI are

$$\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{y'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot(y'-z)\frac{p^+}{k^+}}\,\frac{(y'-z)^m}{(y'-z)^2}\,G_m(y',p^+),$$

where G_m contains $\frac{(x'-y')^{k'}}{(x'-y')^2}$, the other kernels, the bracket of p^+ functions and the colour with $U(y')$.

Lemma (Fourier transform of $\frac{r^m}{r^2}$). For G smooth at $r=0$,

$$\int d^2r\, e^{-iq\cdot r}\,\frac{r^m}{r^2}\,G(r) = -2\pi i\,\frac{q^m}{q^2}\,G(0) + O\Big(\frac{1}{q^2}\Big). \tag{3.4}$$

For a Gaussian G the result is exact: $-2\pi i\frac{q^m}{q^2}(1-e^{-q^2\sigma^2/2})$. Numerically, for $G=e^{-r^2/2}(1+c\cdot r)$, $(1+\zeta)\times$ the integral $/(-2\pi i\frac{k^m}{k^2}G(0)) = 0.999996,\ 1.000015,\ 1.000054,\ 1.000205$ at $q=(1+\zeta)k$, $\zeta=5,10,20,40$.

Consequence. With $q=k(1+\frac{p^+}{k^+})$ and $\int_{w'} e^{-ik\cdot(w'-w)}\frac{(w'-z)^m}{(w'-z)^2} = -2\pi i\, e^{-ik\cdot(z-w)}\frac{k^m}{k^2}$:

$$\int_{y'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot(y'-z)\frac{p^+}{k^+}}\,\frac{(y'-z)^m}{(y'-z)^2}\,G_m(y') \xrightarrow{p^+\gg k^+} \frac{k^+}{p^++k^+}\int_{y'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)^m}{(y'-z)^2}\,G_m(z). \tag{3.5}$$

In the second form y' is a new integration variable, the position of the measured gluon of the conjugate amplitude, and $y'\to z$ everywhere in G_m.

- A term of the p^+ bracket that tends to a constant at large p^+ therefore gives $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+} = \log\frac{\vee}{k^+}$, but only after $\int_{y'}$.
- These terms are $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ of Rows III(1), V, VI and $-\frac{p^+}{k^{+2}}\big[\dots\big]\frac{1}{p^+(x-z)^2+k^+(x-w)^2} \to -\frac{1}{k^{+2}(x-z)^2}\big[\dots\big]$ of Row III part 2.
- No other term gives a logarithm:
  - $\frac{\delta_{k'm}\delta_{ij}}{p^++k^+}$ and the other terms of part 2 fall like $\frac{1}{p^+}$, which with Eq. (3.5) gives $\frac{1}{p^{+2}}$;
  - in $[\frac{1}{p^+}]_+$ only the subtraction $-\frac{f(0)}{p^+}$ (no phase) gives $\log\frac{\vee}{k^+}$, and that term is already in your list;
  - the other singular point $y'=x'$ comes with $e^{-ik\cdot(x'-z)p^+/k^+}$, which oscillates, so it gives no logarithm.

It is the same mechanism as H_4 (the large-momentum tail of $\log(y'-y)^2$ in Group II Row IV). It is also where my earlier statement in `Finite_terms_simplified.pdf` Sec. 5, "the term $e^{-i\kappa\frac{1-a}{a}}$ gives no logarithm after $\int_{y'}$", is wrong: for the t_3 term, $-t_3\frac{1-e^{-i\kappa(1-a)/a}}{i\kappa}$, it gives $-\log\frac{\vee}{k^+}$ times the term below.

The new terms (add to your rows; each with c.c.):

Row III(1):
$$\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&\times\Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big],
\end{aligned} \tag{3.6}$$

Row III(2):
$$\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}\\
&\times\left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2} - \frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]\\
&\Big[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big],
\end{aligned} \tag{3.7}$$

Row V:
$$\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\,\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big],
\end{aligned} \tag{3.8}$$

Row VI:
$$\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big].
\end{aligned} \tag{3.9}$$

- The prefactors are those of the rows: $\frac{ig^4}{4\pi^4}\frac{1}{2\pi}\cdot\frac{1}{k^+}$ for V and VI, the same with $\frac{1}{2}$ for III(1), and $\frac{g^4}{4\pi^4}\frac{1}{2\pi}$ for III(2).
- The sign is that of the $-\frac{1}{k^+}$ and $-\frac{p^+}{k^{+2}}$ in your brackets.
- These are the $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)$ (and, from the c.c., $\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)$) structures of $\mathcal{B}$: the measured gluon is emitted by the soft gluon z.

4 The three-ρ terms cancel

4.1 Rule

For a colour operator with the charges in your order, $O = T^{a_1a_2a_3}\rho^{a_1}(u_1)\rho^{a_2}(u_2)\rho^{a_3}(u_3)$ (anticommutators written out), with $[\rho^a(u),\rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$:

$$O = S(O) + O^{(2)}, \qquad O^{(2)} = \frac{1}{2}\sum_{i<j} O\Big|_{\rho^{a_i}(u_i)\rho^{a_j}(u_j)\to if^{a_ia_jg}\rho^g(u_i)\,\delta(u_i-u_j)}. \tag{4.1}$$

$S(O)$ is the fully symmetric (classical) three-ρ part; $O^{(2)}$ goes into Sec. 5. For classical charges $\{\rho^b(x),\rho^c(y)\} \to 2\rho^b(x)\rho^c(y)$.

4.2 Direct and mirror terms

Every symmetric three-ρ term has one charge with a LO-type factor on one side and two charges on the other.

- **Direct:** one charge x' in $\mathcal{C}^a_L$, two charges x, y on the amplitude side. These are:
  - the Group I rows;
  - Group II Row I part 1 block A;
  - Group III Rows I, II;
  - the *complex conjugates* of Group III Rows III(1), V, VI.
- **Mirror:** all the others. They are the c.c.'s of the direct terms, so it suffices to show that the direct terms cancel.

The rows are brought to the common variables, phase $e^{-ik\cdot(w'-w)}$, by renaming integration variables:

- Group I Row V part 1, first term ($e^{-ik(w'-z)}$): $z\to w$, $x\to z$, $y'\to x$;
- Group I Row V part 1, second term ($e^{-ik(w'-x)}$): $x\to w$, $y'\to x$;
- c.c. of the Group III rows ($e^{+ik(y'-w)}$): $y'\to w$, $w\to w'$, $y\leftrightarrow x'$.

4.3 The structures

Basis: $\Phi[p,q] \equiv f^{abc}\,\mathcal{C}^a_L\,\big(U(p)\rho(x)\big)^b\big(U(q)\rho(y)\big)^c$, with $\big(U(p)\rho(x)\big)^b = U^{bb'}(p)\rho^{b'}(x)$.

- Under the relabelling $x\leftrightarrow y$, $\Phi[p,q]\to-\Phi[\tilde q,\tilde p]$ ($\tilde p$: p with $x\leftrightarrow y$).
- So $\Phi[w,w]$ and $\Phi[x,y]$ are odd and only the antisymmetric part $[\dots]_A$ of their kernels contributes.
- The others pair up: $\Phi[z,y]\simeq-\Phi[x,z]$, $\Phi[w,z]\simeq-\Phi[z,w]$, $\Phi[w,y]\simeq-\Phi[x,w]$, the kernel taken at $x\leftrightarrow y$.

Kernels:

$$K_b = (X'\cdot X_w)K(x,y,z),\quad K_h = (X'\cdot Z_w)K(x,y,z),\quad D = (X'\cdot Y_w)K(x,w,z),\quad E = -(X'\cdot X_w)K(y,w,z),$$

$$K_{\rm IV} = K_h - K_b - E\ \Big(= X'^iY^j\,\mathbb{T}^{ij} - \frac{X'\cdot Y}{2(z-w)^2}\ \text{of your Rows I.IV, II.I(1), III.I, III.II}\Big),$$

$$F_z = (X'\cdot X_w)\big[K(x,y,z)-K(y,w,z)\big] - (X'\cdot Y_w)\big[K(x,y,z)-K(x,w,z)\big],$$

$$\frac{1}{\pi}\int_z F_z = X'\cdot\mathfrak{F}(w;x,y)\ (2.4\times10^{-14}).$$

$\mathfrak{F}^i = \mathcal{K}^i(x-w)\log\frac{(w-y)^2}{(x-y)^2} - \mathcal{K}^i(y-w)\log\frac{(w-x)^2}{(x-y)^2}$ is the kernel of your Group I Row II 2a, written as a z integral.

- $\{\rho(x),\rho(y)\}$ gives $2\rho\rho$.
- $f^{hbc}U^{ah}(p) = f^{ab'c'}U^{b'b}(p)U^{c'c}(p)$.
- The $\mathcal{K}_{\rm IV}$ family and the $\mathcal{K}_C$ family collapse to compact colour factors (`Finite_terms_simplified.pdf`, Eqs. (famIVs), (famCs)).

With these, the symmetric parts of the direct terms are, in units $\frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\,p_8\log\left[\frac{\vee}{k^+}\right]$ (each line checked against the row itself, last column):

row	kernel × symmetric colour	check
I.II 2a	$\frac{1}{2}F_z\big(\Phi[x,y]-\Phi[w,w]\big)$	2.9×10^{-14}
I.III with (A)	$(K_h-K_b)\big(\Phi[x,y]-\Phi[x,z]\big)+D\big(\Phi[x,y]-\Phi[x,w]\big)$	1.0×10^{-15}
I.IV + II.I(1)A + III.I + III.II	$K_{\rm IV}\big(\Phi[x,z]-\Phi[w,w]\big)$	3.0×10^{-15}
I.V(1)	$\frac{1}{2}D\big(\Phi[z,w]-\Phi[x,y]\big)+\frac{1}{2}E\big(\Phi[w,z]-\Phi[x,y]\big)$	1.8×10^{-16}
c.c. of III.III(1) + III.V + III.VI	$-D\big(\Phi[z,w]-\Phi[z,y]-\Phi[x,w]+\Phi[x,y]\big)$	1.5×10^{-14}

Collecting with the pairings above and using $E|_{x\leftrightarrow y}=-D$, $[K_h]_A=0$, $F_z=2[K_b+E]_A$:

$$\begin{aligned}
\Phi[w,w]:&\quad -\tfrac{1}{2}F_z-[K_{\rm IV}]_A=-[K_b+E]_A+[K_b+E]_A=0,\\
\Phi[x,y]:&\quad \big[\tfrac{1}{2}F_z+K_h-K_b+D-\tfrac{1}{2}D-\tfrac{1}{2}E-D\big]_A=\tfrac{1}{2}[E]_A-\tfrac{1}{2}[D]_A=0,\\
\Phi[x,z]:&\quad -(K_h-K_b)+K_{\rm IV}-D|_{x\leftrightarrow y}=-K_h+K_b+K_h-K_b-E+E=0,\\
\Phi[z,w]:&\quad \tfrac{1}{2}D-\tfrac{1}{2}E|_{x\leftrightarrow y}-D=\tfrac{1}{2}D+\tfrac{1}{2}D-D=0,\\
\Phi[x,w]:&\quad -D+D=0.
\end{aligned}$$

All symmetric three-ρ terms cancel, pointwise in z once Row II 2a is written with F_z.

- **Without (A)**, Row I.III has $(X'\cdot Y)K(w,w,z)\big(\Phi[x,y]-\Phi[x,z]\big)$ in place of $D\big(\Phi[x,y]-\Phi[x,w]\big)$. Then $\Phi[x,z]$ is left with $-(X'\cdot Y)K(w,w,z)$ and $\Phi[x,w]$ with $+D$: no cancellation.
- **The terms (B)** have the family colour of the $\mathcal{K}_C$ rows with $y'\to z$, $2f^{abc}[(U(z)-U(x'))\rho(x')]^a[(U(z)-U(x))\rho(x)]^b[(U(w)-U(y))\rho(y)]^c$. It is odd under $x\leftrightarrow x'$, while their kernel $\mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\,K(x',x,z)$ is even. So their symmetric part vanishes by itself (3.4×10^{-17}).

Numerically (engine, all rows, all c.c.'s, the charges placed on three points in all six ways): $|\text{sum}|/\sum|\text{terms}|=3.7\times10^{-17}$, 8.1×10^{-18}, 1.0×10^{-17} (three configurations). As written: 4.5×10^{-2}, 1.2×10^{-2}, 7.9×10^{-4}, 2.2×10^{-3}, 2.7×10^{-2} (five configurations).

5 The two-ρ terms give $\mathcal{B}-\mathcal{J}$

5.1 What is summed

- The commutator terms $O^{(2)}$, Eq. (4.1), of every three-ρ row, in your charge order, with c.c. (reversed order, conjugate coefficient), including $\mathcal{D}$ and the terms (B).
- The genuine two-ρ rows: Group I Row II part 1, Row V 2a, 2b UV, 2b finite; Group II Row I part 2 (2b, 2c), Row IV (both terms); Group III Row III part 2 and its term (B).

5.2 The rows with $\frac{1}{\epsilon}$

In $\overline{\rm MS}$ with $d=2-\epsilon$ (`Part2b_MSbar.pdf`), for $r=|a-b|$,

$$\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2}{4}=\frac{1}{\pi}\int_u\big[K(a,a,u)-K(a,b,u)\big],\qquad \int_u K(a,a,u)=0. \tag{5.1}$$

- **Group I Row II part 1** $=-\mathcal{N}\log\Big[\frac{\vee}{k^+}\Big]\int e^{-ik\cdot(w'-w)}\,N_c\,(X'\cdot X_w)\int_u\big[K(w,w,u)-K(x,w,u)\big]\,\mathcal{C}_L^a\big[U^{ab}(w)-U^{ab}(x)\big]\rho^b(x)$, +c.c.
- **Group I Row V 2b UV** is exactly minus the z integral of the third term of 2b finite, because

  $$\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}=K(z,z,x)-K(y,z,x)\quad(5.2\times10^{-16}).$$

  The two cancel (2b UV is the counterterm of 2b).
- **Group II Row IV.** With $d_\perp=2-\epsilon$,

  $$-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1=-\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}\Big]+O(\epsilon).$$

- **Poles.** The poles of these rows, and of the z integrals of the terms whose colour does not depend on $U(z)$, are kept with the same convention, Eq. (5.1). They match those of $\mathcal{B}-\mathcal{J}$ term by term (the πP rows of Table 2).

5.3 Result, structure by structure

The engine collects all two-ρ terms by kernel monomial, with the two charges at x and x' (both assignments). The colour is evaluated at random points with random classical charges and $U = U^T$.

- Terms whose colour does not depend on $U(z)$ are integrated over z with Eq. (5.1). Then:
  - $\int_z K(a,b,z) = \pi[P - \log(a-b)^2]$, with $P = -\frac{2}{\epsilon} - 2\gamma_E - \log\frac{\mu^2}{4}$;
  - $\int_z K(a,a,z) = 0$.
- The same is done for $\mathcal{B} - \mathcal{J}$, with $\mathcal{B}$, $\mathcal{J}$ parsed from your `Evolution_terms.tex`.
  - Your Sec. 3 equals JIMWLK⊗LO.
  - Both equations of your (commented-out) Sec. 4 equal BFKL⊗LO of App. E.
  - Both comparisons give zero residual in every structure.
- In Table 2, column 3 is the value of $\mathcal{B} - \mathcal{J}$ in that structure at one configuration, and column 4 is the largest relative difference between your rows and $\mathcal{B} - \mathcal{J}$ over three configurations. Row labels: I.III = Group I Row III etc.; * = the terms (B).

Table 2: Two-ρ terms of the corrected list vs. $\mathcal{B} - \mathcal{J}$, kernel structure by kernel structure (the charges at x, x', both assignments).

structure	rows contributing	$\mathcal{B} - \mathcal{J}$	diff.
Pointwise in z			
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(x,w',z)$	I.IV, I.V, II.I(1), III.I, III.II, III.III, III.VI	0	1.7×10^{-16}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(x,x',z)$	I.III, I.IV, III.I, III.II	1.159	1.2×10^{-16}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(x,w,z)$	I.IV, I.V, III.I, III.II, III.III, III.VI, II.I(2)	2.326	1.8×10^{-16}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(x',w',z)$	I.IV, I.V, III.I, III.II, III.III, III.VI, II.I(2)	2.543	9.0×10^{-17}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(x',w,z)$	I.IV, I.V, II.I(1), III.I, III.II, III.III, III.VI	0	1.8×10^{-16}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(w,w',z)$	II.I(2), III.III	-4.391	2.0×10^{-16}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(w',w',z)$	I.V	-1.620	8.4×10^{-17}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(x,x,z)$	I.III, I.IV, II.I(1), III.I, III.II	-6.228	2.5×10^{-16}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(x',x',z)$	I.III, I.IV, II.I(1), III.I, III.II	1.604	3.8×10^{-16}
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)K(w,w,z)$	I.V	4.607	8.8×10^{-17}
$\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)K(x,x',z)$	I.III, I.IV, II.I(1), III.I, III.II, II.I(2), III.III*, III.V*, III.VI*	-0.032	2.3×10^{-16}
$\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)K(x',x',z)$	I.III, I.IV, II.I(1), III.I, III.II, III.III*, III.V*, III.VI*	-3.645	2.9×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)K(x,x',z)$	I.III, I.IV, II.I(1), III.I, III.II, II.I(2), III.III*, III.V*, III.VI*	-1.401	1.5×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)K(x',x',z)$	I.III, I.IV, II.I(1), III.I, III.II, III.III*, III.V*, III.VI*	-14.725	3.9×10^{-17}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')K(x,x',z)$	I.III, I.IV, II.I(1), III.I, III.II	0	1.7×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')K(x',w',z)$	I.IV, I.V, II.I(1), III.I, III.II, III.III, III.VI	0	8.0×10^{-17}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')K(x',w,z)$	I.IV, I.V, II.I(1), III.I, III.II, III.III, III.VI	0	1.4×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(x,w',z)$	I.IV, I.V, III.I, III.II, III.III, III.VI, II.I(2)	0.761	1.1×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(x,x',z)$	I.III, I.IV, III.I, III.II	1.159	3.0×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(x,w,z)$	I.IV, I.V, II.I(1), III.I, III.II, III.III, III.VI	0	1.7×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(x',w',z)$	I.IV, I.V, II.I(1), III.I, III.II, III.III, III.VI	0	1.3×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(x',w,z)$	I.IV, I.V, III.I, III.II, III.III, III.VI, II.I(2)	0.226	1.6×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(w,w',z)$	II.I(2), III.III	1.741	3.8×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(w',w',z)$	I.V	1.491	1.1×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(x,x,z)$	I.III, I.IV, II.I(1), III.I, III.II	-3.039	2.3×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(x',x',z)$	I.III, I.IV, II.I(1), III.I, III.II	0.742	5.5×10^{-17}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')K(w,w,z)$	I.V	-3.082	1.4×10^{-16}
$K(x,w',z)\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	I.IV, I.V, II.I(1), III.I, III.II, III.III, III.VI	0	6.4×10^{-17}
$K(x,x',z)\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)$	I.III, I.IV, II.I(1), III.I, III.II, II.I(2), III.III*, III.V*, III.VI*	-3.788	2.5×10^{-16}
$K(x,x',z)\mathcal{K}(x'-w)\cdot\mathcal{K}(w'-z)$	I.III, I.IV, II.I(1), III.I, III.II, II.I(2), III.III*, III.V*, III.VI*	-2.550	2.6×10^{-16}
$K(x,x',z)\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	I.III, I.IV, II.I(1), III.I, III.II	0	2.2×10^{-16}
$K(x,x',z)K(w,w',z)$	II.I(2), II.IV, III.III*	-7.070	5.0×10^{-16}
$K(x,w,z)\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	I.IV, I.V, II.I(1), III.I, III.II, III.III, III.VI	0	2.8×10^{-16}
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)K(x,x,z)$	I.III, I.IV, II.I(1), III.I, III.II, III.III*, III.V*, III.VI*	-3.574	5.3×10^{-16}

structure	rows contributing	$\mathcal{B}-\mathcal{J}$	diff.
$\mathcal{K}(x'-w)\cdot\mathcal{K}(w'-z)K(x,x,z)$	I.III, I.IV, II.I(1), III.I, III.II, III.III*, III.V*, III.VI*	5.045	1.0×10^{-16}
z-independent colour, integrated over z: logarithms			
$-\pi\log(x-w')^2\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI	0	2.5×10^{-17}
$-\pi\log(x-w')^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, I.V, III.I, III.II, III.III, III.V, III.VI, II.I(2)	7.439	8.2×10^{-16}
$-\pi\log(x-w')^2\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI	0	3.2×10^{-16}
$-\pi\log(x-x')^2\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.II, I.III, I.IV, III.I, III.II	-35.526	2.8×10^{-15}
$-\pi\log(x-x')^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')$	I.II, I.III, I.IV, II.I(1), III.I, III.II	0	5.2×10^{-16}
$-\pi\log(x-x')^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, III.I, III.II	-28.477	1.3×10^{-15}
$-\pi\log(x-x')^2\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, II.I(1), III.I, III.II	0	6.6×10^{-16}
$-\pi\log(x-w)^2\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.II, I.III, I.IV, I.V, III.I, III.II, III.III, III.V, III.VI, II.I(2)	-18.665	1.3×10^{-16}
$-\pi\log(x-w)^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI	0	2.5×10^{-16}
$-\pi\log(x-w)^2\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI	0	2.7×10^{-16}
$-\pi\log(x'-w')^2\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.II, I.III, I.IV, I.V, III.I, III.II, III.III, III.V, III.VI, II.I(2)	-22.477	2.5×10^{-16}
$-\pi\log(x'-w')^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI	0	1.0×10^{-16}
$-\pi\log(x'-w')^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI	0	3.5×10^{-16}
$-\pi\log(x'-w)^2\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI	0	1.3×10^{-16}
$-\pi\log(x'-w)^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI	0	3.0×10^{-16}
$-\pi\log(x'-w)^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, I.V, III.I, III.II, III.III, III.V, III.VI, II.I(2)	3.542	2.8×10^{-16}
$-\pi\log(w-w')^2\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	II.I(2), II.IV, III.III	-23.086	4.5×10^{-16}
$-\pi\log(w-w')^2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	II.I(2), II.IV, III.III	16.068	6.7×10^{-16}
z-independent colour, integrated over z: pole terms			
$\pi P\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI, II.IV	99.754	4.6×10^{-16}
$\pi P\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')$	I.III, I.V, III.III, III.V, III.VI	0	4.7×10^{-16}
$\pi P\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.II, I.III, I.IV, I.V, II.I(1), III.I, III.II, III.III, III.V, III.VI, II.IV	1.429	1.6×10^{-15}
$\pi P\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	I.III, I.V, III.III, III.V, III.VI	0	6.9×10^{-16}
z-independent colour, scaleless ($\int_z K(a,a,z)=0$), compared anyway			
$\int_z K(w',w',z)\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.V	-13.501	1.3×10^{-16}
$\int_z K(w',w',z)\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.V	8.490	1.0×10^{-16}
$\int_z K(x,x,z)\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.III, I.IV, II.I(1), III.I, III.II	-2.375	1.6×10^{-15}
$\int_z K(x,x,z)\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.III, I.IV, II.I(1), III.I, III.II	-8.232	1.3×10^{-15}
$\int_z K(x',x',z)\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.III, I.IV, II.I(1), III.I, III.II	-2.961	1.1×10^{-15}
$\int_z K(x',x',z)\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.III, I.IV, II.I(1), III.I, III.II	-8.718	1.3×10^{-15}
$\int_z K(w,w,z)\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	I.V	-12.916	3.3×10^{-16}
$\int_z K(w,w,z)\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	I.V	8.004	9.3×10^{-17}

All 67 structures agree (largest difference 2.8×10^{-15}; 2 structures are zero on both sides and are not listed). Without (A) and (B), 33 of 75 structures differ; with (A) only, 9 of 67 differ. These are exactly the structures with $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)$, $\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)$ and $K(w,w',z)K(x,x',z)$ that (B) supplies. Hence Eq. (1.1).

6 The BFKL terms

Adding the $\log\frac{\vee}{\Lambda}$ terms of `Evolution_terms.tex` (Sec. 3 there: $\mathcal{N}\log\frac{\vee}{\Lambda}\int\mathcal{J}$) to Eq. (1.1) gives Eq. (1.2):

$$\log\left[\frac{\vee}{\Lambda}\right]\mathcal{J}+\log\left[\frac{\vee}{k^+}\right](\mathcal{B}-\mathcal{J})=\log\left[\frac{k^+}{\Lambda}\right]\mathcal{J}+\log\left[\frac{\vee}{k^+}\right]\mathcal{B}.$$

The BFKL term is therefore $\mathcal{N}\log\frac{\vee}{k^+}\int\mathcal{B}$. In your notation (your Sec. 4, second equation, with $\log\frac{\vee}{k^+}$):

$$\frac{d^3N_{\mathrm{BFKL}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\Bigg[$$

$$
\begin{aligned}
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\Big[\\
&- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\Big[\\
&- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
&+ 2\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) - 2\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&- 2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) + 2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\Big]\\
&+ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
&+ 2\,N_c\,\rho^a(x)\,\rho^a(x') - \ f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')\\
&- \ f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)\\
&+ \ f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x)\,\rho^e(x') + \ f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x')\,\rho^e(x)\\
&+ \ f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x') + \ f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)\Big]\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\Big[\\
&- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\Big]\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
&+ 2\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') - 2\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')\\
&- 2\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') + 2\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\Big[\\
&- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')\\
&+ N_c\,U^{ab}(z)\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') - N_c\,U^{ab}(z)\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
&+ 2\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) - 2\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)\\
&- 2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) + 2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)\Big]\Big]
\end{aligned}
\tag{6.1}
$$

- One by one, the $\frac{1}{(x-z)^2}$ and $\frac{1}{(x'-z)^2}$ terms are not integrable at $z\to x$, $z\to x'$.
- In $\mathcal{B}$ these singularities cancel against the $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)$ and $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)$ terms at the same points. In the singlet form, Eq. (7.1) below, lines 3 and 8 cancel at $z\to y$, and so do lines 2 and 6 after $x\leftrightarrow y$.
- In Sec. 5 the terms with z-independent colour are treated in dimensional regularization, as your $\frac{1}{\epsilon}$ rows are.

The JIMWLK companion, $\mathcal{N}\log\frac{k^+}{\Lambda}\int\mathcal{J}$, is your Sec. 3 with $\log\frac{\vee}{\Lambda}\to\log\frac{k^+}{\Lambda}$.

7 Colour-singlet form

Projectile in a colour singlet: $\rho^b(x')\rho^c(x)\to\delta^{bc}\frac{\langle\rho^f(x)\rho^f(y)\rangle}{N_c^2-1}$ with $x'\to y$; $U^\dagger=U^T$.

- The forms below are your singlet forms of `Evolution_terms.tex`, identical to the corrected ones I checked before (`Evolution_terms_corrected.pdf`, Eqs. (3.4), (4.3)).

- They equal the singlet projection of $\mathcal{J}$ and $\mathcal{B}$, and App. E, Eq. (LOBFKL), to machine precision (`../check_v4/check_singlet.json`, `../bfkl/bfkl_singlet.json`).
- Only the logarithms are new, and $\int_{x,,z,w,w'} \to \int_{x,y,z,w,w'}$ (blue).

BFKL.

$$\begin{aligned}
\frac{d^3N_{\rm BFKL}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\frac{1}{N_c^2-1}\log\Big[\frac{\vee}{k^+}\Big]\int_{x,y,z,w,w'} e^{-ik(w'-w)}\,\langle\rho^f(x)\rho^f(y)\rangle\,\Bigg[\\
\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
+K(w,w',z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}\Big]\\
+\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big]\\
+\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]\Bigg]
\end{aligned}\tag{7.1}$$

JIMWLK companion.

$$\begin{aligned}
\frac{d^3N_{\rm JIMWLK}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\frac{1}{N_c^2-1}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,y,z,w,w'} e^{-ik(w'-w)}\,\langle\rho^f(x)\rho^f(y)\rangle\,\Bigg[\\
\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[4N_c{\rm Tr}\{U^\dagger(w)U(w')\}\\
-2{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}-2{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-4N_c{\rm Tr}\{U^\dagger(x)U(w')\}\\
+2{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}+2{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[4N_c{\rm Tr}\{U^\dagger(x)U(y)\}\\
-2{\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\}-2{\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-4N_c{\rm Tr}\{U^\dagger(w)U(y)\}\\
+2{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}+2{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-2N_c{\rm Tr}\{U^\dagger(w)U(w')\}+2N_c{\rm Tr}\{U^\dagger(x)U(w')\}\\
-{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(w')U(z)\}-{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\\
+{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}+{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w')\}\\
+{\rm Tr}\{T^eU^\dagger(w')U(x)\}\,{\rm Tr}\{T^eU^\dagger(w')U(z)\}+{\rm Tr}\{T^eU^\dagger(w)U(w')\}\,{\rm Tr}\{T^eU^\dagger(w')U(z)\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[-2N_c{\rm Tr}\{U^\dagger(w)U(w')\}+2N_c{\rm Tr}\{U^\dagger(w)U(y)\}\\
+{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w)U(z)\}+{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\\
-{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(w)U(z)\}-{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\\
-{\rm Tr}\{T^eU^\dagger(w)U(w')\}\,{\rm Tr}\{T^eU^\dagger(w)U(z)\}+{\rm Tr}\{T^eU^\dagger(w)U(y)\}\,{\rm Tr}\{T^eU^\dagger(w)U(z)\}\Big]\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[2N_c{\rm Tr}\{U^\dagger(x)U(w')\}-2N_c{\rm Tr}\{U^\dagger(x)U(y)\}\\
-{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}-{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}
\end{aligned}$$

$$
\begin{aligned}
&+\operatorname{Tr}\{T^e U^\dagger(x)U(y)T^e U^\dagger(x)U(z)\}+\operatorname{Tr}\{T^e U^\dagger(x)U(y)T^e U^\dagger(z)U(x)\}\\
&-\operatorname{Tr}\{T^e U^\dagger(w')U(x)\}\operatorname{Tr}\{T^e U^\dagger(x)U(z)\}-\operatorname{Tr}\{T^e U^\dagger(x)U(y)\}\operatorname{Tr}\{T^e U^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[2N_c\operatorname{Tr}\{U^\dagger(w)U(y)\}-2N_c\operatorname{Tr}\{U^\dagger(x)U(y)\}\\
&-\operatorname{Tr}\{T^e U^\dagger(w)U(y)T^e U^\dagger(y)U(z)\}-\operatorname{Tr}\{T^e U^\dagger(w)U(y)T^e U^\dagger(z)U(y)\}\\
&+\operatorname{Tr}\{T^e U^\dagger(x)U(y)T^e U^\dagger(y)U(z)\}+\operatorname{Tr}\{T^e U^\dagger(x)U(y)T^e U^\dagger(z)U(y)\}\\
&-\operatorname{Tr}\{T^e U^\dagger(w)U(y)\}\operatorname{Tr}\{T^e U^\dagger(y)U(z)\}+\operatorname{Tr}\{T^e U^\dagger(x)U(y)\}\operatorname{Tr}\{T^e U^\dagger(y)U(z)\}\Big]\Big]
\end{aligned}
\tag{7.2}
$$

Total (Eq. (1.2) in singlet form): $\frac{d^3N_{\rm LL}}{d^3k}$ = (the JIMWLK line with log $\frac{k^+}{\Lambda}$) + (the BFKL line with log $\frac{\vee}{k^+}$). At mid rapidity, $k^+=\sqrt{\vee\Lambda}$, both logarithms are $\frac{1}{2}\log\frac{\vee}{\Lambda}$.

8 Corrections to my earlier notes

- **`BFKL_terms_check.pdf` and `Finite_terms_1_1_corrected.pdf`.** I wrote that, with the exact coefficients of `Finite_terms_1.tex`, the three-ρ terms cancel and the two-ρ terms give log $\left[\frac{\vee}{k^+}\right](\mathcal{B}-\mathcal{J})$.
  - That is true only with (A) and (B).
  - I had checked one structure (C_w) directly and relied on `LogVeeKplus_BFKL.pdf`, which has $\mathcal{D}$ and the region $y'\to z$ (its region-P rows). The rows of `Finite_terms_1.tex` have neither.
  - The corrected list there lacks the same two changes.
- `Finite_terms_split.pdf`. The BFKL section (Secs. 2–4) lacks (A) and (B). In Sec. 4, "the other terms of these rows have no log $\frac{\vee}{k^+}$" is wrong. The remaining terms of Group III Rows III, V, VI are therefore not finite as written: the terms (B), times -1, must be taken out of them (moved to BFKL). The same holds for the edge-kernel terms of Group I Row III.
- `Finite_terms_simplified.pdf` (Sec. 5, Group III Rows III, V, VI): the $e^{-i\kappa(1-a)/a}$ part of the t_3 term does give log $\frac{\vee}{k^+}$ after $\int_{y'}$ (Sec. 3.2).
- `Finite_terms_DGLAP_RC.pdf`. The DGLAP and running-coupling terms are unaffected. The finite terms of Group III Rows III, V, VI still contain the terms (B), and Group I Row III its edge kernel; both move as above.

9 Files

In `nlo_gluon_paper/bfkl_final/`:

`input/bfkl.tex`	your file as received
`bfkl_corrected.tex/.pdf`	your file with (A) and (B) in red/blue
`rows.py`	every row in the reordering engine (your charge order, c.c.)
`validate.py`	engine vs. numerical transcription (Table 1)
`three_rho_test.py`, `three_rho_engine.py`	three-ρ cancellation
`phi_table.py`	the Φ table of Sec. 4, row by row
`two_rho_engine.py`, `yz_test.py`, `table2.py`	two-ρ terms vs. $\mathcal{B}-\mathcal{J}$ (Table 2)
`rowcmp.py`	row by row vs. `../bfkl` (the region-P analysis)
`checks.py`	identities: Eq. (3.1), F_z, (B) three-ρ, the lemma, the counterterm
`make_corrected.py`, `make_note.py`, `static.tex`	the two documents